The jth block total - $x_{.j} = \sum_{i=1}^{k} x_{ij}$, $j = 1, 2, ..., m$

Note that, $k \times m = n$, where m = number of blocks, and k = number of treatments (groups) and n is the total number of observations in the study.

The total variation present in the observations x_{ij} can be split into the following three components:

i) The variation between treatments (groups)

ii) The variation between blocks.

iii) The variation inherent within a particular setting or combination of treatment and block.

3.4.1 Test Procedure

Steps involved in two-way ANOVA are:

Step 1 : In two-way ANOVA we have two pairs of hypotheses, one for treatments and one for the blocks.

Framing Hypotheses

Null Hypotheses

H_{01}: There is no significant difference among the population means of different groups (Treatments)

H_{02}: There is no significant difference among the population means of different Blocks

Alternative Hypotheses

H_{11}: Atleast one pair of treatment means differs significantly

H_{12}: Atleast one pair of block means differs significantly

Step 2 : **Data** is presented in a rectangular table form as described in the previous section.

Step 3 : **Level of significance** α.

Step 4 : **Test Statistic**

$$F_{0t}(\text{treatments}) = \frac{MST}{MSE}$$

$$F_{0b}(\text{block}) = \frac{MSB}{MSE}$$

To find the test statistic we have to find the following intermediate values.

i) Correction Factor: $C.F = \frac{G^2}{n}$ where $G = \sum_{j=1}^{m}\sum_{i=1}^{k} x_{ij}$

ii) Total Sum of Squares: $TSS = \sum_{i=1}^{k}\sum_{j=1}^{m} x_{ij}^2 - C.F$

iii) Sum of Squares between Treatments: $SST = \sum_{i=1}^{k} \frac{x_{i.}^2}{m} - C.F$

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